

Ying Xu

North China Electric Power University, Beijing, China
fibrecase@ncepu.edu.cn · fibrecase.github.io · github.com/FibreCase

EDUCATION

North China Electric Power University

2023.09 – 2027.06 (expected)

B.Eng. in New Energy Science and Engineering (wind-energy focus)

- GPA: 3.29/5.00; calculated weighted average: 82.9/100.
- Selected coursework: Wind Resources Measurement and Assessment (95/100), Numerical Simulation (90/100), Reliability Engineering (90/100), Wind Turbine Aerodynamics (Excellent).

RESEARCH INTERESTS

Wind-energy integration; wind-turbine modeling and dynamics; renewable power systems.

RESEARCH EXPERIENCE

Multi-Timescale Modeling of PMSG Wind Turbines

2026.09 – present

Undergraduate research project, NCEPU · Advisor: Dong Zhao

- Reviewed 12 papers on wind-turbine models, model reduction, voltage ride-through, and frequency response; developed an initial framework separating mechanical, power-control, and current-control dynamics.
- Current focus: model formulation and parameter specification for a direct-drive permanent-magnet synchronous generator (PMSG) turbine; planned evaluation covers wind-speed, voltage, and frequency disturbances.

CFD Studies of Airfoil Aerodynamics and Vortex-Induced Vibration

2026.07 – 2026.08

Wind Power Research Center, NCEPU · Supervised by a doctoral researcher

- Completed five simulation studies under guidance: cylinder flow and vibration, NACA0012 static and dynamic stall, and S809 dynamic stall; used ANSYS Fluent and ran a cylinder-flow case in OpenFOAM.
- Compared simulated lift and pitching-moment hysteresis with experimental reference data for NACA0012 dynamic stall; documented cylinder-vibration responses and flow-field evolution in a technical presentation.

AWARDS & HONORS

- **National First Prize**, China Undergraduate Electronic Design Contest (2025); nominated for the TI Cup
- National Third Prize, China Undergraduate Electronic Design Contest — TI Cup Analog Electronic System Design (2024)
- Beijing First Prize, Undergraduate Electronic Design Contest (2024)
- University Merit Student; Second-Class University Scholarship; Science and Technology Innovation Scholarship

SELECTED ENGINEERING PROJECTS

WindAssess — Wind-Resource Assessment Toolkit

2026.06

- Built a Python pipeline for SODAR/LIDAR and met-mast wind data, with range, correlation, trend, and missing-data checks; implemented Weibull fitting, wind-shear estimation, air-density calculations, and wind roses, with unit tests.

Monocular-Vision Target Measurement Device

2025.07 – 2025.08

National First Prize team · Hardware and system integration

- Designed the power-monitoring circuit, adapted the MIPI display, and integrated hardware and software on an embedded Linux platform during a four-day competition.
- The team achieved target-dimension errors below 2 mm at approximately 1.6 m; measured system power was 7 W peak and 5 W idle, within the 10 W limit.

12 V Power Monitor

2026.09 – present

- Built a custom PCB and ESP32-C3 firmware for current/power measurement, with hardware and software overcurrent protection; developed a Python backend for data logging and live visualization.

NanoScope — Two-Channel Pocket Oscilloscope

2025.06 – 2025.07

- Built a dual-channel instrument with an analog front end, STM32 ADC/DMA acquisition, and ESP32 display; implemented frequency/phase measurement over 50 Hz–200 kHz and Bluetooth control.

LEADERSHIP

Head of Technology Department, MOS Innovation Lab, NCEPU

2025.05 – present

- Organized STM32 training and mentored new members; designed technical assessments and led project reviews for dozens of members.

TECHNICAL SKILLS

Simulation ANSYS Fluent (2D external flow); OpenFOAM (basic)

Programming Python, C; MATLAB (basic)

Engineering STM32 / ESP32, PCB design, embedded Linux; Git, Docker, SolidWorks

Languages English: IELTS Academic 7.0 (September 2026)